HUANYU CABLE | POWERING PROJECTS SINCE 1995

TECHNICAL CATALOG

SOLAR PHOTOVOLTAIC CABLES

H1Z2Z2-K | PV1-F

Single-core flexible solar cable solutions with tinned copper conductors for photovoltaic DC systems.

REFERENCE STANDARDS

EN 50618:2014

2 PfG 1169/08.2007

Hefei Huanyu Wire & Cable Co., Ltd.

www.huanyucable.com

PORTFOLIO OVERVIEW

Solar Cable Solutions

A focused product range for DC-side connections in photovoltaic systems.

HUANYU CABLE supplies solar cable solutions for international projects, with product construction, marking, documents and packing confirmed against the applicable standard and project requirement.

Representative product rendering - final marking and construction subject to confirmation.

Typical PV DC connection path

01	02	03	04
PV Modules	**String Circuit**	**Combiner Box**	**Inverter**
Module interconnection	String wiring	DC collection	DC input connection

01 Tinned Copper

Class 5 flexible stranded conductor with a tin-plated surface for corrosion resistance.

02 Halogen-free

Cross-linked insulation and outer sheath designed for solar cable service.

03 Outdoor Service

Resistance to UV, ozone, moisture and weathering under the applicable standard.

04 Red / Black

Color options support positive and negative DC circuit identification.

Product images are representative renderings. Final construction, marking, certificate coverage and packing are subject to order confirmation.

PRODUCT 01

H1Z2Z2-K Solar Cable

Single-core flexible cable for the DC side of photovoltaic systems.

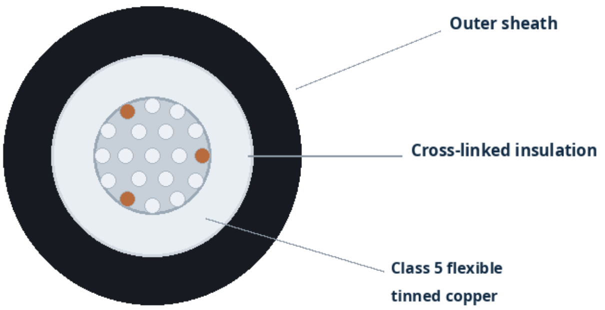
EN 50618:2014

H1Z2Z2-K

Designed for long-term indoor and outdoor PV DC connections, including module strings, combiner boxes and inverter circuits.

RATED DC VOLTAGE **1.5/1.5 kV**	RATED AC VOLTAGE **1.0/1.0 kV**
NORMAL CONDUCTOR **90 °C**	SHORT CIRCUIT **250 °C / 5 s**

Cable construction



Key technical characteristics

- Class 5 flexible stranded tinned copper conductor
- Cross-linked halogen-free insulation and outer sheath
- Maximum operating DC voltage up to 1.8 kV
- Up to 120 °C for a limited cumulative period of 20,000 h
- Design life basis of at least 25 years under specified conditions

Common requested sizes 1×2.5 | 1×4 | 1×6 | 1×10 | 1×16 mm²

Other sizes subject to technical and commercial confirmation.

Technical values are summarized from EN 50618:2014 for preliminary product selection.

Performance Overview

Key standard-based characteristics for technical discussion.

01 Electrical Performance

1.5/1.5 kV DC rating, 1.0/1.0 kV AC rating and a maximum operating DC voltage not exceeding 1.8 kV under the standard.

02 Thermal Performance

Normal maximum conductor temperature of 90 °C; limited operation at 120 °C; short-circuit conductor temperature of 250 °C for 5 seconds.

03 Mechanical Performance

Flexible Class 5 conductor construction with cold bend, cold elongation, impact, shrinkage and dynamic penetration test requirements.

04 Environmental Resistance

UV weathering, ozone resistance, damp heat, moisture and outdoor service requirements for photovoltaic installations.

05 Fire & Material Behaviour

Halogen-free materials, low-smoke performance and vertical flame propagation testing under the applicable referenced methods.

06 Marking & Traceability

Designation, nominal size and manufacturer / traceability information are normally repeated along the cable sheath.

Typical application conditions

Indoor / outdoor	Fixed PV DC installations
Solar modules	Module and string wiring
Combiner boxes	DC collection circuits
Inverters	DC input connections

Actual ampacity, dimensions, bending radius and packing depend on the confirmed product design and installation conditions.

PRODUCT 02

PV1-F Solar Cable

For projects and markets that specifically request the PV1-F designation.

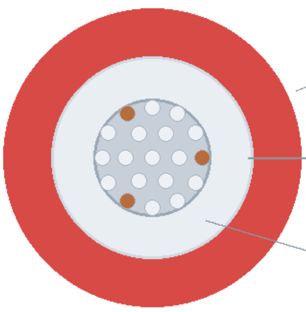
2 PfG 1169/08.2007

PV1-F

Flexible, halogen-free solar cable design for photovoltaic DC circuits where the PV1-F designation is specified by the customer or project.

RATED AC VOLTAGE **0.6/1 kV**	DC SYSTEM USE **Up to 1.8 kV**
OPERATING RANGE **-40 to +90 °C**	SHORT CIRCUIT **200 °C / 5 s**

Cable construction



Key technical characteristics

- Class 5 flexible stranded tinned copper conductor
- Halogen-free cross-linked insulation and sheath design
- Maximum conductor temperature of 120 °C under specified conditions
- Resistance to UV, ozone, weathering and damp-heat exposure
- 25-year design life basis under specified operating conditions

Common requested sizes 1×2.5 | 1×4 | 1×6 | 1×10 | 1×16 mm²

Other sizes subject to technical and commercial confirmation.

Technical values are summarized from 2 PfG 1169/08.2007 for preliminary product selection.

PV1-F

Performance Overview

A concise technical summary for project inquiry and preliminary selection.

01 Electrical Performance

0.6/1 kV AC rated design with DC application conditions up to 1.8 kV, subject to the specification and system voltage relationship.

02 Thermal Performance

Operating temperature range from -40 °C to +90 °C, conductor temperature up to 120 °C and short-circuit temperature of 200 °C for 5 seconds.

03 Mechanical Performance

Flexible conductor construction with low-temperature impact, bend, elongation, dynamic penetration and cut-propagation test requirements.

04 Environmental Resistance

UV weathering, ozone, damp heat, acid/alkali exposure and moisture-related tests for outdoor photovoltaic use.

05 Fire & Chemical Behaviour

Flame propagation, smoke density, halogen content, pH, conductivity and toxicity-related requirements are included in the specification.

06 Project Compatibility

Final acceptance should follow the buyer specification, local market requirements and the certificate / test document scope requested by the project.

Ordering formats

Cable colour	Black / red
Packing	Coil or drum
Length	Confirmed per order
Marking	Confirmed per order

Certification logos and certificate numbers are intentionally omitted until the applicable product and brand scope is confirmed.

PRODUCT SELECTION

H1Z2Z2-K or PV1-F?

Select the designation according to the buyer specification, market practice and required compliance documents.

Selection item	H1Z2Z2-K	PV1-F
Reference document	EN 50618:2014	2 PfG 1169/08.2007
Primary voltage expression	1.5/1.5 kV DC	0.6/1 kV AC; DC use conditions
Short-circuit temperature	250 °C / 5 s	200 °C / 5 s
Typical inquiry situation	New EN 50618-based PV projects	Projects specifically requesting PV1-F
Selection principle	Confirm project standard and certificate scope	Confirm project acceptance and document scope

Information required for quotation

01 **Model / standard** H1Z2Z2-K, PV1-F or project specification

02 **Cable size** Nominal cross-section in mm²

03 **Quantity** Total length and preferred coil / drum length

04 **Colour** Black, red or project-specific requirement

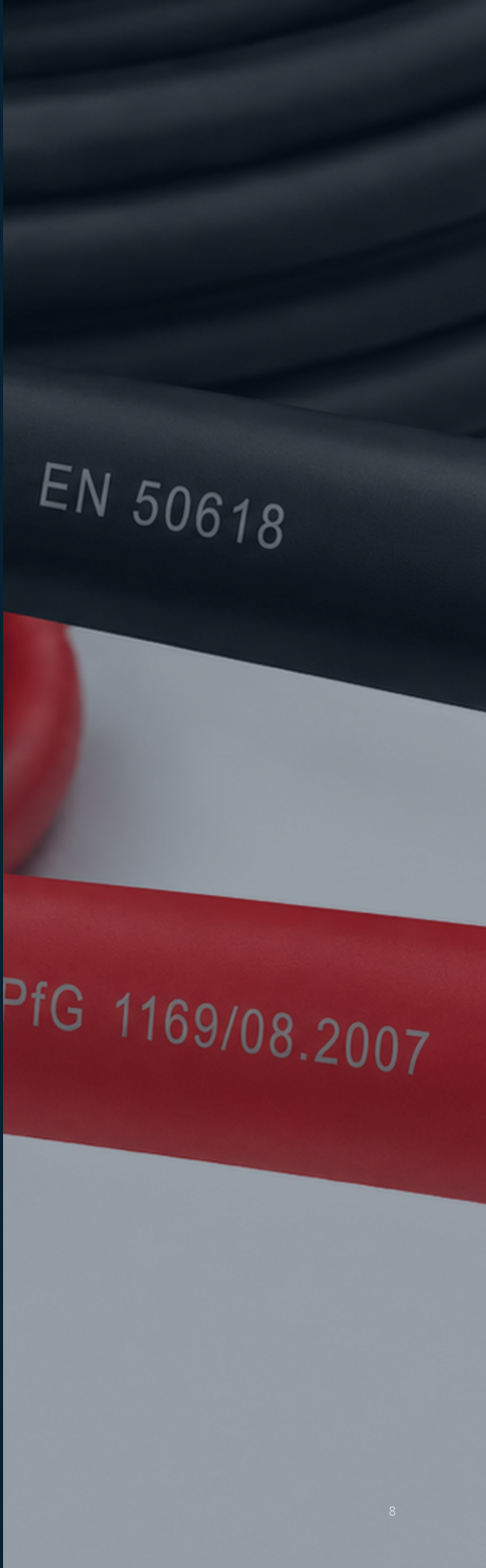
05 **Destination** Country, port and requested trade term

06 **Documents** Certificate, test report or local approval requirement

Final product selection and compliance documentation must follow the confirmed purchase specification and destination-market requirements.

CABLE SOLUTIONS FOR GLOBAL PROJECTS

Founded in 1995, Hefei Huanyu Wire & Cable Co., Ltd. focuses on low- and medium-voltage power cables and project-oriented cable supply. Solar photovoltaic cables are offered as a complementary product line for international customers.



Project support

- Specification and model review
- Preliminary quotation and order coordination
- Document and certification scope confirmation
- Export packing, marking and delivery coordination

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Technical note This catalog provides preliminary technical reference only. Final construction, dimensions, marking, certification, test documentation and packing are subject to written order confirmation.